Journal of Equine Veterinary Science 102 (2021) 103620

Contents lists available at ScienceDirect

Journal of Equine Veterinary Science

journal homepage: www.j-evs.com

Histidine-Rich Glycoprotein Functions as a Dual Regulator of Neutrophil Activity in Horses

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ARTICLE INFO

Article history:
Received 24 February 2021
Received in revised form 30 March 2021
Accepted 6 April 2021
Available online 16 April 2021

Keywords:
Horses
Histidine-rich glycoprotein
Neutrophils
Reactive oxygen species
Systemic inflammatory response syndrome

ABSTRACT

Histidine-rich glycoprotein (HRG) is an abundant plasma protein that has been identified in most mammals. We first identified whole genome sequence of equine *HRG* (e*HRG*) and succeeded to purify eHRG from plasma of horses. Since HRG interacts with various ligands, this protein is thought to be involved in immune response, coagulation, and angiogenesis. Systemic inflammatory response syndrome (SIRS) is characterized as a non-specific, clinical, pro-inflammatory immune response that damage organs and tissues in the host. Recent reports revealed that blood HRG levels in human patients with SIRS are approximately 50% lower than those in healthy controls, indicating the use of HRG as a biomarker or treatment for SIRS. SIRS is also a serious issue in equine medicine. In this study, we investigated various effects of eHRG on neutrophil functions, including adhesion, migration, phagocytosis, reactive oxygen species (ROS) production, and lysosome maturation using neutrophils isolated from horses. Microscopic observation showed that the addition of eHRG to the culture diminished adhesion of neutrophils stimulated with LPS. Using the Boyden chamber technique, we showed that eHRG reduced neutrophil chemotaxis induced by recombinant human IL-8. Luminol-dependent chemiluminescence assay demonstrated that eHRG restrained the peak of LPS-promoted ROS production from neutrophils. In contrast, eHRG promoted phagocytic activity evaluated with uptake of fluorescent dye conjugated particles, as well as lysosomal maturation assessed using fluorescent staining for lysosomes of equine neutrophils. These results indicated that eHRG acts as a dual regulator of neutrophils in horses.

1. Introduction

Histidine-rich glycoprotein (HRG) is an abundant plasma glycoprotein that interacts with many types of molecules, including heparin, phospholipids, plasminogen, fibrinogen, immunoglobulin G, C1q, heme, and zinc ions [1–5]. Past observations have indicated that HRG is involved in immune response, coagulation, and angiogenesis. Recently, HRG has been reported to affect neutrophil adhesion, morphology, and reactive oxygen species (ROS) production from neutrophils of mice and humans [6]. Human neutrophils incubated with HRG maintained a spherical shape compared to those incubated without HRG, indicating that HRG has suppressive effects on neutrophils. Recently, we demonstrated the whole se-

Conflict of Interest: The authors declare no potential conflicts of interest. None of the authors has any financial or personal relationships that could inappropriately influence or bias the content of the article.

Animal welfare/ethical statement: The experimental protocol was approved by the standards specified in the guidelines provided by the Animal Care and Use Committee of the Tokyo University of Agriculture and Technology as well as those in the Science Council of Japan's guidelines for the use of experimental animals (approval nos. 30-104 and R02-46).

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https://doi.org/10.1016/j.jevs.2021.103620

quence of equine HRG (eHRG) and isolated eHRG from the serum of horses [7]. In addition, we determined that eHRG is produced by the liver in horses, similar to mice and humans [7]. However, the role of eHRG in infections and/or inflammatory conditions in horses remains unclear.

Equine systemic inflammatory response syndrome (SIRS), which often develops in horses with colic and colitis, was first defined by referring to the definition of human SIRS [8, 9]. In a recent equine medicine, SIRS criteria in foals and adult horses have been presented [8–10]. Particularly in adult horses, intestinal diseases complicated with SIRS would be associated with a high risk of death [8]. The rate of SIRS horses transferred to the ICU is 31.3%, and the case fatality rate of SIRS horses is 38.8% [8]. However, the mechanisms that initiate a series of processes resulting in SIRS remain uncertain. Moreover, a sensitive biomarker and an effective treatment protocol for equine SIRS have not yet been determined.

The recent study revealed the 50% decline of plasma HRG in human patients with SIRS comparing with healthy subjects and suggested the possibility of HRG as a new biomarker for SIRS [11]. In addition, HRG-deficient mice were more susceptible to *Candida parapsilosis* and *Streptococcus pyogenes* [12,13], indicating the protective role of HRG against infectious pathogens. In fact, administration of HRG to septic mice in a cecal ligation and puncture model improved their viability [6]. These results raise the possibility that HRG constitutes a novel strategy for the early diagnosis and treatment of SIRS.

In this study, we examined the effects of eHRG on equine neutrophils to investigate the role of eHRG in inflammatory conditions. We found that eHRG suppressed lipopolysaccharide (LPS)-stimulated adhesion and ROS production in neutrophils. In addition, eHRG reduced interleukin (IL)-8-induced migration of equine neutrophils. In contrast, eHRG enhanced LPS-activated phagocytosis and lysosome maturation in equine neutrophils. Here, we demonstrated for the first time that eHRG affects the regulation of neutrophil functions in horses.

2. Material and Methods

2.1. Isolation of Equine Neutrophils

Equine blood was collected from the jugular vein using heparinized tubes. Collected blood was centrifuged at 1,500 rpm for 15 min, and the lower fraction containing the buffy coat and red blood cells was layered on top of Percoll (GE Healthcare, Piscataway, NJ, USA) layers (70% and 85%) [14,15]. After density gradient centrifugation at 400 $\times$ *g* for 20 min, the cells between the 70% and 85% layers were collected. The collected cells underwent Giemsa staining to confirm the purity of the neutrophils and a trypan blue dye exclusion test to determine their viability. Neutrophils were resuspended in Roswell Park Memorial Institute (RPMI)-1640 medium (Life Technologies, Gaithersburg, MD, USA) and stored at room temperature with rotation. The purity of neutrophils determined by Giemsa staining was $>$ 95%. We collected samples from 2 healthy warm-blooded horses (KWPN and Westfalen, geldings, 18 and 19 years old) according to the standards specified in the guidelines provided by the Animal Care and Use Committee of the Tokyo University of Agriculture and Technology as well as those in the Science Council of Japan's guidelines for the use of experimental animals (approval nos. 30-104 and R02-46).

2.2. HRG Purification from Equine Plasma

eHRG was purified using equine plasma as described previously [7,16]. Briefly, equine plasma was incubated with Ni Sepharose 6 Fast Flow (GE Healthcare, Piscataway, NJ) for 10 min at room temperature. After incubation, a flow-through solution was collected, and the sepharose was washed with 50 mM Tris-HCl (pH 7.5) containing 0.5 M NaCl and 20 mM imidazole. A flow-through solution was stored and used as a control. The binding eHRG was then eluted with 50 mM Tris-HCl (pH 7.5) containing 0.5 M NaCl and 500 mM imidazole. The elution fraction was further purified by centrifugation using an Amicon Ultra 0.5 mL 30 kDa centrifugation filter (Millipore, Bedford, MA, USA). We have been creating the quantitative ELISA method for eHRG in equine plasma. From our preliminary data, eHRG concentrations in the plasma of healthy horses would be 1–10 μM. According to the results, we set eHRG concentrations 1–10 μM in all experiments.

2.3. Morphology and Adhesion Assay

To analyze neutrophilic morphology in the presence of eHRG, purified equine neutrophils resuspended in RPMI-1640 medium with calcein-AM (Dojindo, Kumamoto, Japan) were seeded at a density of 1 $\times$ 10^5 cells/well in a 96-well plate with or without 5 μM eHRG and incubated at 37°C and 5% CO_2 for 1 h. We used 56.7 μg/mL flow-through solution (56.7 μg/mL: the same protein concentration of 1 μM eHRG) as a control. After incubation, the neutrophil shape was assessed using a Keyence BZ-X810 fluorescence microscope (Osaka, Japan).

The adhesion assay was performed using the described method with a slight modification [6]. Purified equine neutrophils (5 $\times$ 10^4 cells/well) were suspended in a 96-well plate in culture medium (RPMI-1640 medium containing 10% fetal bovine serum, 100 U/mL penicillin, 100 μg/mL streptomycin, and 2 mM GlutaMAX [Thermo Fisher Scientific, Waltham, MA, USA]) with calcein-AM in the presence or absence of 1, 5, or 10 μM purified eHRG or 56.7 μg/mL flow-through solution for 1 h at 37°C and 5% CO_2. After incubation, 50 ng/mL LPS (Sigma-Aldrich, St. Louis, MO, USA) was added to each well, and the cells were incubated for 1 h. Then, each well was washed lightly with 100 μL culture medium, and the number of adherent neutrophils with fluorescence was counted using a Keyence BZ-X810 fluorescence microscope. Obtained data from six independent experiments was used for statistical analysis.

2.4. Chemotaxis Assay

The chemotactic activity of IL-8-stimulated neutrophils was measured using the ChemoTx Chemotaxis System (Neuro Probe, Gaithersburg, MD, USA) with a 3 μm pore size [17–19]. Culture medium with or without 5 ng/mL recombinant human IL-8 (rhIL-8; R&D Systems, Minneapolis, MN, USA) and 1, 5, or 10 μM purified eHRG or flow-through solution was added to the bottom chamber. As a negative control, 1 μg/mL anti-human IL-8 antibody was added to the bottom wells. Neutrophils stained with calcein-AM were applied to the upper part above the filter (2 $\times$ 10^4 cells/well) in the presence of each concentration of eHRG or flow-through solution, and the plate was incubated at 37°C and 5% CO_2 for 1 h. After incubation, the solution in the upper part was removed, and the cells in the bottom chamber were collected via trypsin-ethylenediaminetetraacetic acid treatment. Then, the fluorescence of neutrophils that migrated to the bottom chamber was counted using a Keyence BZ-X810 fluorescence microscope. The proportion of pHrodo-positive area (red) in Calcein-AM-positive area (green) was calculated using the Keyence Hybrid Cell Count application. Calculated data was shown as the ratio to the negative control. Obtained data from three independent experiments was used for statistical analysis.

2.5. Phagocytosis Assay

The phagocytic activity of neutrophils was measured using pHrodo *Escherichia coli* BioParticles conjugate (Thermo Fisher Scientific) according to the manufacturer's protocol. Neutrophils were resuspended in Hank's Balanced Salt Solution (HBSS) with calcein-AM in the presence or absence of various concentrations of eHRG or 56.7 μg/mL flow-through solution and seeded into each well of a 96-well plate (1×10^5 cells/well), which was then incubated at 37°C with 5% CO_2 for 1 h. For opsonization, pHrodo *E. coli* BioParticles were incubated with 50% horse serum in HBSS at 37°C with 5% CO_2 for 30 min. Then, pHrodo *E. coli* BioParticles were collected by centrifugation and resuspended in HBSS. The opsonized pHrodo *E. coli* BioParticles were added to each well, and the plate was incubated for 1 h. The plate was subsequently centrifuged at $400 \times g$ for 5 min, and the fluorescence of phagocytosed particles was observed using a Keyence BZ-X810 fluorescence microscope. The proportion of pHrodo-positive area (red) in Calcein-AM-positive area (green) was calculated using the Keyence Hybrid Cell Count application. Obtained data from six independent experiments was used for statistical analysis.

2.6. Luminol-Dependent Chemiluminescence Assay

ROS production by neutrophils was measured using a described method with a slight modification [19,20]. Equine neutrophils in culture medium were seeded into a 96-well plate (2×10^5 cells/well) and incubated in the presence or absence of 1 μM eHRG or flow-through solution for 1 h. After incubation, 100 μM luminol (Sigma-Aldrich) was added with or without 50 ng/mL LPS and/or 1 μM eHRG and 56.7 μg/mL flow-through solution, and chemiluminescence was measured every minute using a Mithras LB 940 microplate reader (Berthold Technologies, Bad Wildbad, Germany). Obtained data from three independent experiments was used for statistical analysis.

2.7. Fluorescence Analysis of Lysosomal Acidification

Lysosomal acidification of equine neutrophils was analyzed using LysoTracker Red DND-99 (Thermo Fisher Scientific) according to the manufacturer's protocol. Briefly, purified equine neutrophils were incubated in RPMI-1640 medium with calcein-AM and 1 μM LysoTracker in the presence or absence of the indicated concentration of eHRG or flow-through solution. After incubation for 1 h at 37°C with 5% CO_2, 50 ng/mL LPS was added to the medium, and the cells were cultured for 30 min. Then, the fluorescence of acidic lysosomes was observed using a Keyence BZ-X810 fluorescence microscope. The proportion of LysoTracker-positive area (red) in Calcein-AM-positive area (green) was calculated using the Keyence Hybrid Cell Count application. Calculated data was shown as the ratio to controls. Obtained data from three independent experiments was used for statistical analysis.

2.8. Statistical Analysis

The data of Luminol-dependent chemiluminescence assay were compared by one-way ANOVA, followed by Newman-Keuls test. Tukey's test was performed for the statical analysis of the data in other experiments. All analyses were performed using JMP software (version 14.0.0). P values of < 0.05 were considered statistically significant.

3. Results

3.1. Effects of eHRG on Morphology and Adhesiveness of Neutrophils

First, we examined the effect of eHRG on the morphology of equine neutrophils. Non-adherent spherical cells show clear round shape, whereas adherent cells are obscure in their outlines with ameboid shapes. Neutrophils stimulated with LPS showed adhesive morphology and the number of adherent cells was about 6-fold higher than that of the control without LPS (Fig. 1A and B). In contrast, the addition of eHRG completely suppressed the adhesion of equine neutrophils induced by LPS, the number of LPS stimulated adherent cells incubated with 1 μM eHRG was 62% lower compared with the LPS-stimulated control (Fig. 1A and B). Cells incubated with eHRG showed spherical shapes with or without LPS stimulation (Fig. 1A). The flow-through solution, which we used as another control, did not suppress the adhesion of stimulated neutrophils. Consequently, purified eHRG retained neutrophils in a spherical shape, even when the cells were stimulated with LPS.

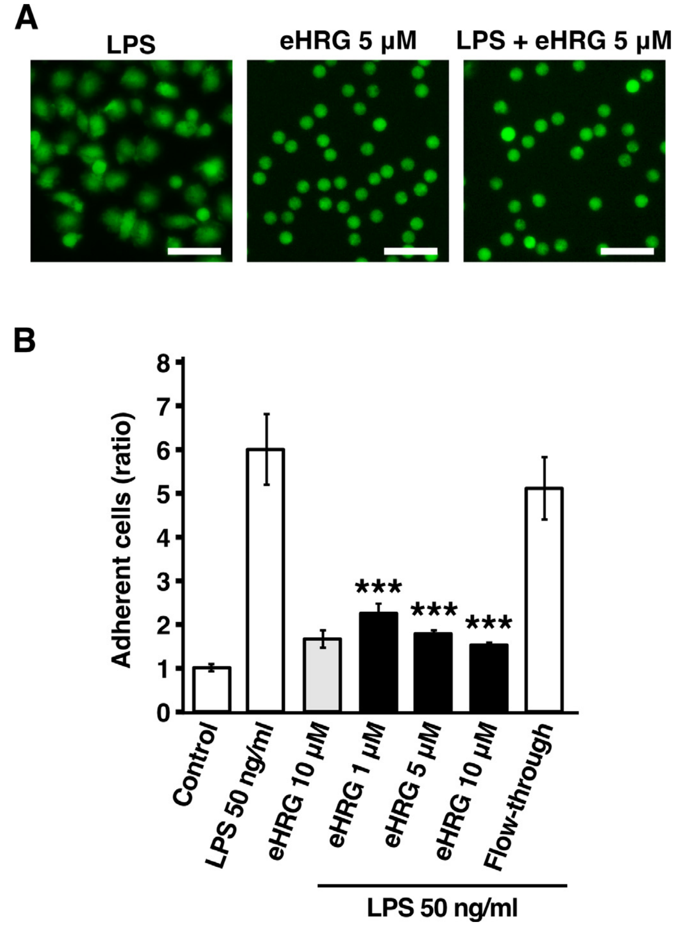

Fig. 1. (A) Morphology of equine neutrophils in the presence of equine histidine-rich glycoprotein (eHRG). Purified neutrophils labelled with calcein-AM were cultured with or without 50 ng/mL lipopolysaccharide (LPS) and/or 5 μM eHRG for 1 h, and cell shape was observed using a fluorescence microscope (scale bars: 30 μm). (B) The adhesiveness of equine neutrophils in the presence of eHRG with LPS stimulation. Equine neutrophils were precultured with or without 1, 5, or 10 μM eHRG or flow-through solution for 1 h. After incubation, the cells were stimulated with 50 ng/mL LPS for 1 h. Then, each well was washed, and adherent neutrophils were counted using the Keyence Hybrid Cell Count application. Calculated data was shown as the ratio to controls. Data represent the means ± standard errors of six independent experiments. $^{***}P < 0.001$ compared to the LPS-treated control using Tukey's test.

3.2. Turn-Down Effect of eHRG on IL-8-Induced Chemotaxis of Neutrophils

The effects of eHRG on cell migration were examined using rhIL-8 and purified equine neutrophils. Flow-through solution was used as the negative control. As shown in Fig. 2A and B, migration of neutrophils was induced by rhIL-8 and completely inhibited by the anti-hIL-8 antibody, indicating the chemotactic activity of rhIL-

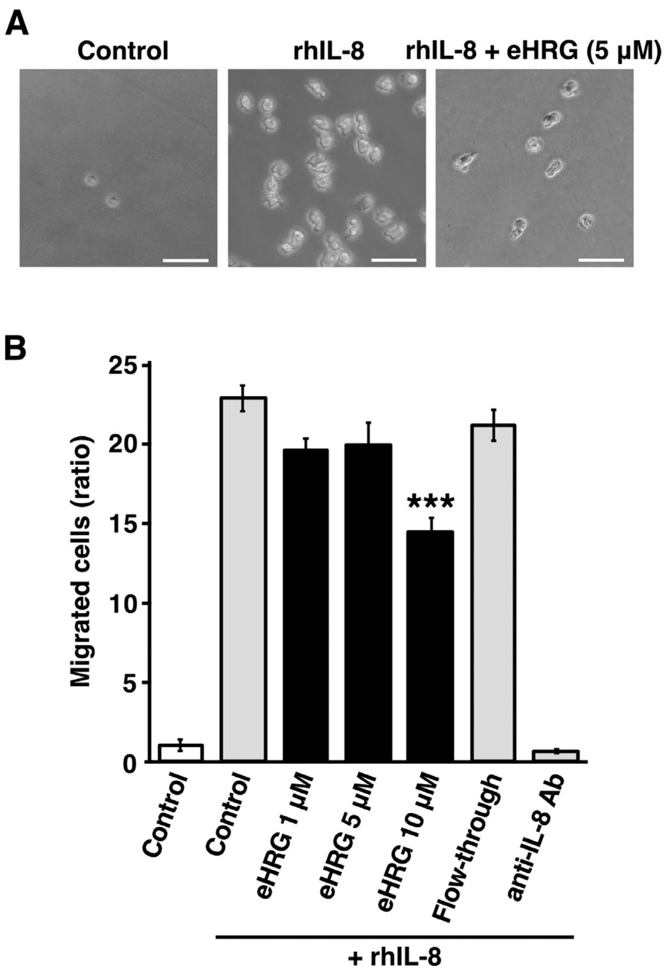

Fig. 2. (A) Migrated neutrophils in the presence of equine histidine-rich glycoprotein (eHRG). Chemotaxis of equine neutrophils was induced by 5 ng/mL human recombinant interleukin (IL)-8 with or without 1, 5, or 10 μM eHRG or flow-through solution for 1 h, and cells that migrated to the bottom chamber were observed using a fluorescence microscope. Next, 1 μg/mL anti-human IL-8 antibody was used for confirming IL-8-dependent chemotaxis (scale bars: 30 μm). (B) Quantification of migrated neutrophils. Migrated cells were counted using the Keyence Hybrid Cell Count application. Calculated data was shown as the ratio to controls. Data represent the means ± standard errors of three independent experiments. $^{***}P < 0.001$ compared to the rhIL-8-induced control using Tukey's test.

8 on equine neutrophils. The number of migrated cells induced by rhIL-8 was about 23-fold higher compared with the negative control without rhIL-8. Cells that migrated to the bottom chamber were significantly decreased in the presence of 10 μM eHRG compared to the rhIL-8-treated control. The suppression rate of 10 μM eHRG treatment to rhIL-8-induced neutrophil migration was about 37%. No suppressive effect of the flow-through solution on neutrophil chemotaxis was observed. These results indicate that a high concentration of eHRG attenuates rhIL-8-induced neutrophil chemotaxis.

3.3. Upregulation of Phagocytic Activity of Neutrophils by eHRG

Neutrophilic phagocytosis in the presence or absence of eHRG was measured using the pHrodo *E. coli* BioParticles conjugate. The conjugate was generated from *E. coli* and used to induce the phagocytic activity of cells. Equine neutrophils showed an adhesive shape when cultured with pHrodo and little uptake of the particles (Fig. 3A and B). Most particles of pHrodo were observed outside of each cell (Fig. 3A). When equine neutrophils were cultured with purified eHRG, the uptake of pHrodo particles was markedly

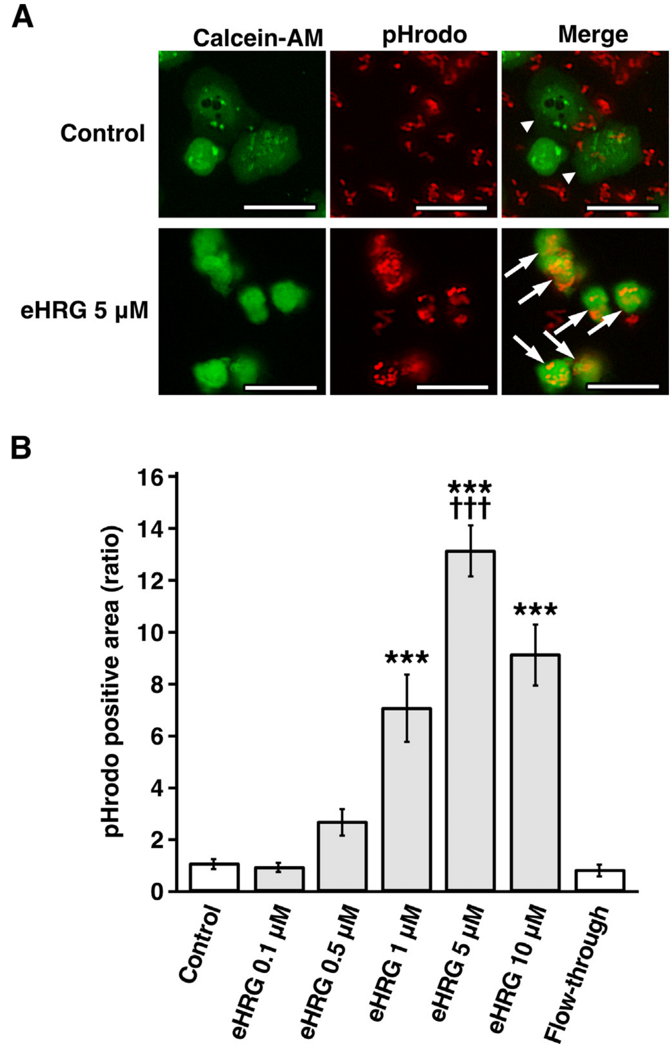

Fig. 3. (A) Phagocytosis of equine neutrophils with equine histidine-rich glycoprotein (eHRG). Purified neutrophils were cultured in Hank's Balanced Salt Solution with or without 0.1, 0.5, 1, 5, or 10 μM eHRG or flow-through solution for 1 h. Opsonized pHrodo *Escherichia coli* BioParticles with horse serum were added to each well, and the cells were incubated for 1 h. The fluorescence of phagocytosed particles was observed using a fluorescence microscope (scale bars: 20 μm). Arrow heads show neutrophils with adhesive morphology. Arrows show neutrophils that contain pHrodo particles. (B) Quantification of the fluorescence intensity of phagocytosed pHrodo *E. coli* BioParticles. The proportion of pHrodo-positive area (red) in Calcein-AM-positive area (green) was calculated using the Keyence Hybrid Cell Count application. Calculated data was shown as the ratio to controls. Data represent the means ± standard errors of six independent experiments. $^{***}P < 0.001$ compared to the control and $^{\dagger\dagger\dagger}P < 0.001$ compared to 1 μM eHRG using Tukey's test.

promoted in a dose-dependent manner; the uptake of pHrodo particles was about 7-fold higher with 1 μM eHRG and was about 13-fold higher with 5 μM eHRG than the control (Fig. 3A and B). pHrodo particles were found inside of each cell (Fig. 3A). In contrast, supplementation with flow-through solution did not induce the uptake of pHrodo particles.

3.4. Suppressive Effects of eHRG on the ROS Production of Neutrophils

Since neutrophils stimulated with LPS intensively produce ROS, we investigated the effects of eHRG on ROS production using neutrophils isolated from horses. ROS production from equine neutrophils was increased by LPS stimulation (Fig. 4A and B). In contrast, in the presence of eHRG, the peak of ROS production by LPS stimulation was significantly suppressed (Fig. 4A and B). The sup-

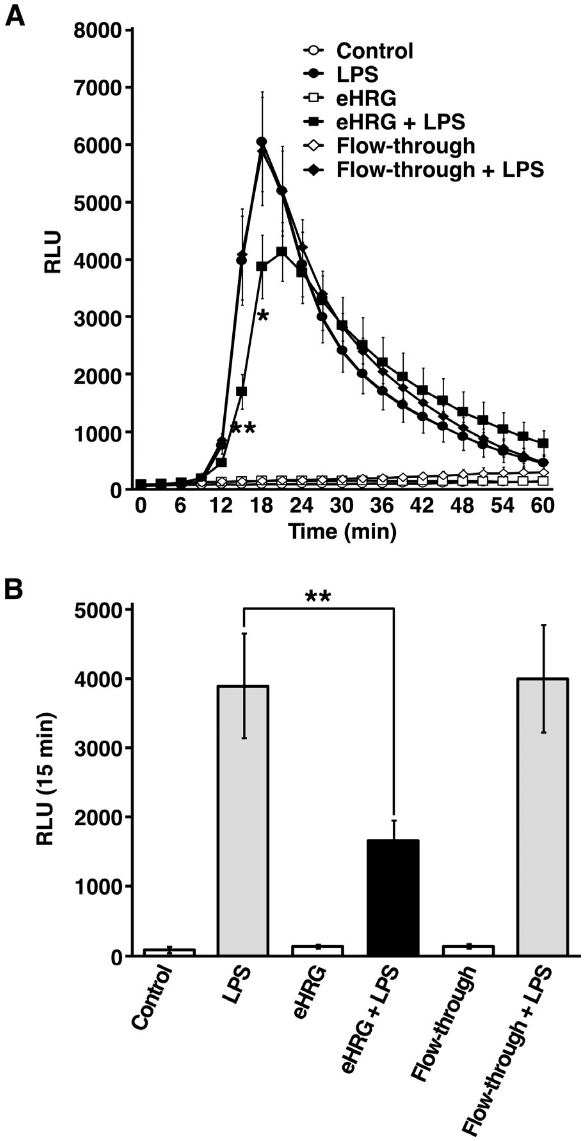

Fig. 4. (A) Reactive oxygen species (ROS) production in the presence of equine histidine-rich glycoprotein (eHRG). Purified neutrophils were incubated with or without 1 μM eHRG or flow-through solution for 1 h before stimulation. Then, the cells were stimulated with 50 ng/mL lipopolysaccharide (LPS), and ROS production was detected using the luminol-dependent chemiluminescence assay. Data represent the means ± standard errors of three independent experiments. $^{*}P < 0.05$ and $^{**}P < 0.01$ compared to LPS using Newman-Keuls test. (B) Quantification of ROS production. The relative light unit (RLU) value as ROS production at 15 min was displayed. Data represent the means ± standard errors of three independent experiments. $^{**}P < 0.01$ compared to ROS production of LPS-stimulated cells using Newman-Keuls test.

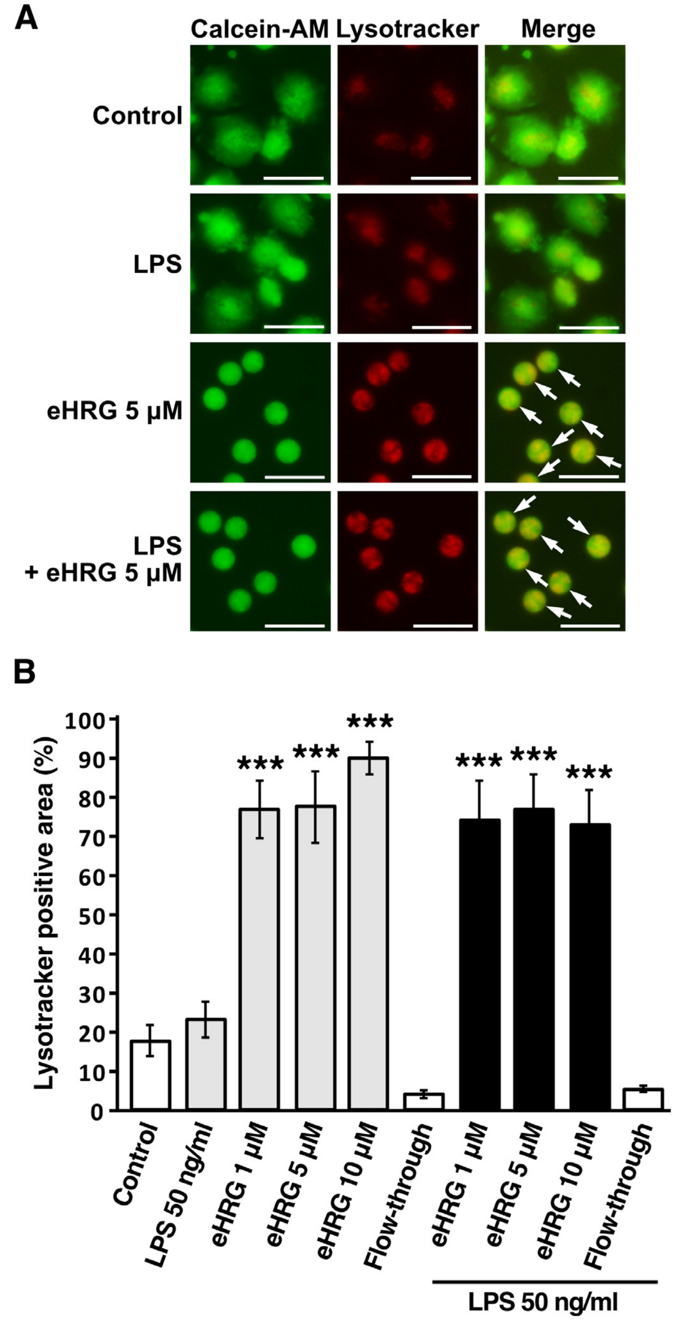

Fig. 5. (A) Lysosome maturation in the presence of equine histidine-rich glycoprotein (eHRG). Purified neutrophils were incubated with LysoTracker in the presence or absence of 1, 5, or 10 μM eHRG or flow-through solution for 1 h. Then, the cells were stimulated with 50 ng/mL lipopolysaccharide (LPS) and incubated for 30 min. The fluorescence of acidic lysosomes was observed using a fluorescence microscope (scale bars: 20 μm). Arrows show neutrophils that contain acidic lysosomes. (B) Quantification of the area of acidic lysosomes. The proportion of LysoTracker-positive area (red) in Calcein-AM-positive area (green) in each cell was calculated using the Keyence Hybrid Cell Count application. Calculated data was shown as the percentage of LysoTracker-positive area in Calcein-AM-positive area in each cell. Data represent the means ± standard errors of three independent experiments. $^{***}P < 0.001$ compared to the control using Tukey's test.

pression rate was 25%. Flow-through solution did not affect LPS-induced ROS production (Fig. 4A and B).

3.5. Promotion of Lysosomal Acidification by eHRG

We found that eHRG upregulated the phagocytic activity of equine neutrophils (Fig. 3). After engulfment of the pathogen, the phagosomes fuse with preformed lysosomes containing many kinds of enzymes that have optimal pH levels for their activation [21–23]. Therefore, lysosomal pH is important for maturation of lysosomes leading to the elimination of target particles. Thus, we examined the effects of eHRG on lysosome maturation by visualizing the pH change of cytosolic lysosomes in neutrophils. Lysosomal acidification of neutrophils in the presence of purified eHRG was evaluated using LysoTracker to confirm the biosynthesis of acidic lysosomes. When equine neutrophils were incubated in the presence of eHRG, the accumulation of acidic lysosomes in neutrophils was markedly increased compared to that in the controls (Fig. 5A and B). Interestingly, the addition of LPS had no influence on the lysosome acidification of neutrophils. Flow-through solution had no effect on lysosomal acidification.

4. Discussion

In this study, we first revealed the effects of eHRG on adhesiveness, chemotaxis, phagocytosis, and lysosomal acidification of neutrophils isolated from the peripheral blood of horses. In some experiments, we used LPS as a stimulant for equine neutrophils, since horses are quite sensitive to LPS and LPS-mediated conditions have become a serious problem in horse medicine.

LPS induced morphological changes in equine neutrophils, in contrast, neutrophils maintained round shape by the addition of eHRG. These results indicate that eHRG may prevent the adhesion of neutrophils by keeping spherical conditions. Since adhesive morphological changes in neutrophils are associated with extravasation and neutrophils show adhesive behavior at the affected sites of bacterial infection, eHRG may function as a downregulator of neutrophil activation at this point.

IL-8 is an intensive chemokine that initiates neutrophil migration. Infected and damaged tissues promote IL-8 production and neutrophil infiltration to the affected sites [24,25]. Migration of equine neutrophils was also induced by rhIL-8, and high concentrations of eHRG restrained this peak. On the other hand, 1 μM human HRG did not inhibited chemotaxis of human neutrophils induced by N-formyl-methionyl-leucyl-phenylalanine (fMLP) [6]. Further examination may differentiate effects of eHRG on neutrophil chemotaxis to various chemoattractants.

ROS are known to react with DNA, lipids, proteins, and enzymes, resulting in gene mutations, protein degeneration, and decreases in enzymatic activity [24]. Bacterial components, such as LPS, are potent stimulators that induce ROS production, which exacerbates the clinical conditions of patients with SIRS [26–28]. In human neutrophils, extracellular and intracellular ROS production in quiescent state was suppressed in the presence of 1 μM human HRG [6]. In this study, eHRG reduced the peak of ROS production of LPS-stimulated neutrophils, suggesting that eHRG may regulate over-production of ROS from neutrophils in damaged tissues.

Interestingly, eHRG increased pHrodo particle uptake, showing that eHRG may upregulate the phagocytic activity of equine neutrophils. Moreover, since eHRG promoted the accumulation of acidic lysosomes in the cytosol of neutrophils without LPS, eHRG may support neutrophilic lysosomal acidification in circulation. These results delineate the various effects of eHRG on neutrophil functions. Reagents used for systemic inflammation should be immunoregulatory but not immunosuppressive because some patients suffer from infectious diseases. In this study, we observed regulatory effects of eHRG on neutrophils, indicating that eHRG may support horses in recovering from serious systemic inflammation, such as SIRS.

Our hypothesis on the biological functions of eHRG *in vivo* is shown in Fig. 6. When thrombi are formed by pathological factors, thrombin activity is increased in the microenvironment. Since HRG is degraded by thrombin [6], the concentration of eHRG will be decreased compared to that in circulation in such a situation. A decrease in eHRG is associated with the adhesiveness of neutrophils to the vascular endothelium, initiating extravasation and chemotaxis to infected sites, promoting bactericidal action by ROS production. On the other hand, at a physiological concentration of eHRG in circulation, adhesiveness, chemotaxis, and ROS production will be suppressed, whereas phagocytic ability is sufficiently maintained. Lysosomal vesicles associated with endocytosis may be matured in the cytosol of neutrophils under physiological eHRG levels in circulation. This scheme suggests that HRG functions as a switch of neutrophilic activity according to its gradient in circulation and in damaged endothelium and tissues. How eHRG controls these neutrophilic activities remains unclear; however, eHRG may become a novel molecule that we focus on for diagnosis and treatment of equine SIRS.

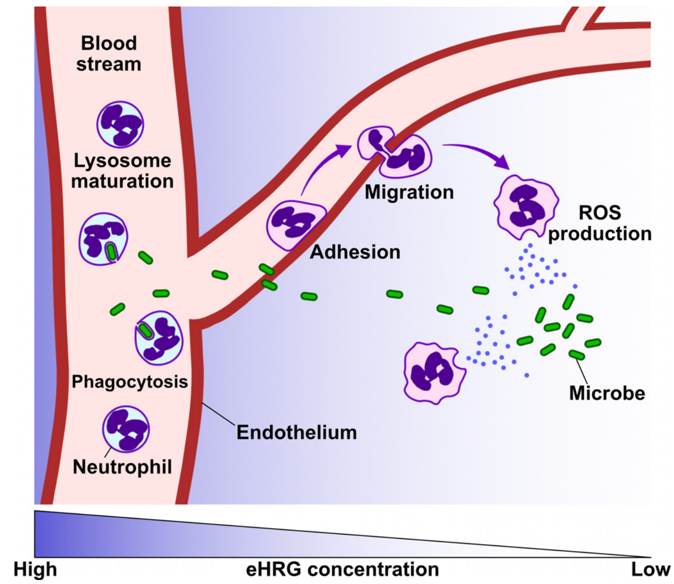

Fig. 6. Scheme of neutrophilic activity in the concentration gradient of equine histidine-rich glycoprotein (eHRG).

5. Conclusions

In this report, we first isolated eHRG and examined its effects on equine neutrophils. We found that eHRG suppressed LPS-stimulated adhesion of neutrophils and maintained their round shape. LPS-induced ROS production by neutrophils was suppressed by treatment with eHRG. In addition, eHRG reduced the chemotaxis of equine neutrophils to rhIL-8. In contrast, eHRG enhanced phagocytosis and lysosomal acidification in equine neutrophils. Here, we demonstrated for the first time that HRG functions as a possible switch of neutrophilic activity according to its gradient.

Author Contributions

Ryo Muko: Methodology, Validation, Formal analysis, Investigation, Writing – Original Draft, Visualisation.
Hiroshi Matsuda: Conceptualization, Validation, Writing – Review & Editing, Supervision.
Masa-aki Oikawa: Validation, Writing – Review & Editing.
Taekyun Shin: Validation, Writing – Review & Editing.
Kenshiro Matsuda: Formal analysis, Writing – Review & Editing.
Hiroaki Sato: Validation, Writing – Review & Editing.
Tomoya Sunouchi: Formal analysis, Validation, Writing – Review & Editing.
Akane Tanaka: Conceptualization, Methodology, Validation, Writing – Review & Editing, Supervision, Project administration.

Funding

This work was supported by the Japan Society for the Promotion of Science KAKENHI [grant numbers JP16H06383, JP19H00969, 20J13519] and a grant for race-horse production and development 2019 and 2020 provided by the Japan Racing Horse Association.

Acknowledgements

We thank Ms. Naomi Noguchi and Ms. Sayaka Maeno (Tokyo University of Agriculture and Technology) for their excellent horse care as well as Ms. Mayuko Nagasaki, Ms. Mayumi Shibayama, and Dr. Kaoru Karasawa (Tokyo University of Agriculture and Technology) for their technical and medical support.

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